Bridge report: crypto mathematics (T3, HQH-539-512, Π) ↔ RNT / RTA / g_k / g_s / Bridge C

For Arvin Hampton (539 Labs) · 2026-10-23, about 10:06 PM CDT. Folder `/workspace/quantum-comp/reports/bridges/CRYPTO_RNT_RTA/` (new files only; nothing pushed, sent or published; CORE_FREEZE untouched).

Conclusion

No bridge exists yet. The only shared object is the map T3 itself, which appears with the same definition in the hash and in the RTA manuscript; that is a common definition, not a link. Three preregistered candidate maps were tested and all three were falsified: T3 orbits do not prefer squares mod k, the fibre sizes 67/68 do not single out g_67 or g_68, and T3 does not preserve the fibre blocks. Three more candidates (539/s = 7 structure, T3 as an RTA operation, the 3-adic measure vs OccFilt) give no testable prediction and stay analogy only or not definable. New Category A facts about the crypto map alone: T3 on Z_3 is measure-isomorphic to the uniform 3-shift (P2), and T3^L sends uniform residues mod 3^L·M to uniform residues mod M for every M coprime to 6 (P3). The fibre blocks 68/68/68/67×5 are exactly the balanced 8-part partition of 539 (P1), so 67 and 68 are determined by 539 and the choice of 8 blocks. They carry no extra information. The Residual-vs-Resonant lock stays in force. Nothing here says that number-theory structure adds security to the hash.

Candidate maps

Map	Definition (short)	Prediction	Status
M1	share of T3-orbit values in squares mod k	excess ≥ 0.02 at ≥ 3 of k = 7, 11, 17, 67	**falsified** (1 of 4; signs mixed)
M2	block size b ∈ {67, 68} ↦ zero count of g_b	g_67, g_68 extreme among k = 60..76	**falsified** (67 is 4th, 68 middle); numerology
M3	T3 on fibre blocks mod 539	diagonal ≥ 1.5× independence	**falsified**, exact (1.108, 1.072, 1.004 for L = 1, 2, 3)
M4	539 = 7²·11, 18+521 vs s = 7 Bridge C / ρ_s level	none definable	**analogy only** (numerological; "68·7+67" = 543 ≠ 539)
M5	T3 as an RTA operation	none (RTA has no algebraic axioms; same object)	**not a bridge**
M6	3-adic measure of T3 vs OccFilt_k	none (n mod k not continuous on Z_3)	**analogy only**; produced the proved facts P2, P3 about T3

The 67/68 section

- Category A (P1): 539 = 8·67 + 3, so the unique 8-part partition with parts differing by at most 1 has 3 parts of 68 and 5 of 67. RTA manuscript Props 4.2–4.3 state the same.
- No source says why there are 8 blocks. The one mod-8 remark (`Object539_NonCircular_Construction.md:222`) is marked "Heuristic only".
- Tested consequence (M2): falsified. The zero counts follow the arithmetic type of k (primes and 2·primes have many zeros), as the known zero families predict, and not the fibre sizes.
- Data on the open class k ≡ 0 mod 4 (no verdict): g_68 has its last zero at d = 4064 within d ≤ 20000. The late zeros lie in classes {8, 32, 52, 60, 64} mod 68.
- Incidental: g_7(67) = +1 (ledger entry 90) and 68 = 4·17. These are coincidences of small integers, and no map was built on them.

What a proved bridge would need

1. A map defined before testing from T3-side objects (Z_3 dynamics, residue classes, itineraries) into g_k / ρ_s objects that is functorial or equivariant. An example is an $SL_2(Z)$- or Hecke-compatible operator built from T3. P3 shows that residue-class statistics of T3 are uniform, so any link has to come from finer structure such as itinerary words or 3-adic expansions, not from occupation densities.
2. A definition of the residual predicate R and the permanent class on a path (still undefined in the sources). Without it, the residual objects cannot enter Π.
3. A proof, not a correlation. Under the lock, a bridge counts only if it predicts a new statement on the other side that is then proved.

Recommended next step

Choose whether to pursue the itinerary route. Define, before any data, a map from T3 itinerary words of length j (uniform by P2) to weights on residues mod k, and ask whether it reproduces OccFilt_k(∞) = squares mod k for odd k. If you do not want this, record the bridge program as "no link found; T3 facts P1–P3 added" and keep the crypto and number-theory lines separate.

Decisions for Arvin

1. Whether 8 blocks (hence 67/68) has any intended source. If not, treat the fibre sizes as fixed by 539 alone and drop 67/68 as a link target.
2. Whether to pursue the itinerary-word map above (preregistered), or close the bridge program as negative.
3. Whether P2 and P3 should go into the Category A ledger as T3 entries (they are proved here and depend on nothing else).

Files: INVENTORY.md, CANDIDATE_MAPS.md, PREREG.md, PREREG.sha256, RESULTS.md, NOTES.md, scripts/, logs/.